

1.3 Classwork: Arithmetic Series — Markscheme

1. [Maximum mark: 5]

The series $3 + 7 + 11 + 15 + \dots$, so $u_1 = 3$ and $d = 4$.

(a) $S_5 = 3 + 7 + 11 + 15 + 19 = 55$ **A1**

Accept 55 obtained by any method, including a sum formula.

The sum of only four terms, $3 + 7 + 11 + 15 = 36$, earns no mark.

(b) $S_5 = \frac{5}{2}(2(3) + (5 - 1)(4))$ **M1**

$S_5 = \frac{5}{2}(22) = 55$ **A1**

Accept substitution into the second formula, $S_5 = \frac{5}{2}(3 + 19) = 55$.

M1 is for substituting $n = 5$, $u_1 = 3$ and $d = 4$ (or $u_5 = 19$), however the arithmetic is then arranged. The question asks for the substituted values, so 55 with no substitution shown earns the A1 only.

(c) $S_{20} = \frac{20}{2}(2(3) + (20 - 1)(4))$ **M1**

$S_{20} = 10(6 + 76) = 820$ **A1**

Accept the route $u_{20} = 3 + 19(4) = 79$, then $S_{20} = \frac{20}{2}(3 + 79) = 820$.

Using nd in place of $(n - 1)d$, giving $10(6 + 80) = 860$, earns the M1 only.

Accept 820 from a GDC sum of the first 20 terms for full marks. Feedback: write out the formula substitution so the method is visible.

2. [Maximum mark: 7]

$u_1 = 18$, $d = 3$, and there are 20 rows.

$$\text{(a)} \quad u_{20} = 18 + (20 - 1)(3) \quad \text{M1}$$

$$u_{20} = 75 \text{ seats} \quad \text{A1}$$

Using 20 in place of $(n - 1)$, giving $18 + 20(3) = 78$, earns the M1 only.

$$\text{(b)} \quad S_{20} = \frac{20}{2}(18 + 75) \quad \text{M1}$$

$$S_{20} = 10(93) = 930 \text{ seats} \quad \text{A1}$$

$$\text{Accept } S_{20} = \frac{20}{2}(2(18) + 19(3)) = 930.$$

FT: when $S_n = \frac{n}{2}(u_1 + u_n)$ is used, award M1A1 for the total correctly obtained from the candidate's own u_{20} in (a).

(c) Seats in the first 8 rows:

$$S_8 = \frac{8}{2}(2(18) + (8 - 1)(3)) \quad \text{M1}$$

$$S_8 = 4(57) = 228 \quad \text{A1}$$

$$\text{Seats not reserved: } 930 - 228 = 702 \quad \text{A1}$$

Accept the direct route: rows 9 to 20 are 12 rows from $u_9 = 42$ to $u_{20} = 75$, so $\frac{12}{2}(42 + 75) = 702$. Award M1 for a sum of these 12 rows, A1 for $u_9 = 42$, A1 for 702.

Summing the first 9 rows in place of the first 8 ($S_9 = 270$, answer 660) earns the M1 and the final A1 by FT, but not the A1 for 228.

FT: award the final A1 for the candidate's own total from (b) minus the candidate's own reserved-row total.